Space-time approach to Electromagnetism

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Abstract

Classical electrodynamics is presented from a relativistic point of view, then gauge theories and general spacetimes are discussed.

I. INTRODUCTION

Relativity emerged from the study of Maxwell's equations. In the early 1900s, a major concern to physicists was that mechanics seemed to satisfy the principle of relativity, that the laws of physics should take the same form in all inertial reference frames; it seemed that electrodynamics did not. The resolution would come with the advent of the special theory of relativity. Steven Weinberg in *Gravitation and Cosmology*[2] writes,

> "...the principle of relativity was not originated by the special theory of relativity, but *restored* by it. Before Maxwell, it might have been supposed that all of physics is invariant under the Galileo group. Maxwell's equations were not invariant under this group, and for half a century it appeared that only mechanics, not electrodynamics, obey the principle of relativity. After Einstein, it was clear that the equations of both mechanics and electrodynamics are invariant, but with respect to the Lorentz group, not the Galileo group"

It should then come as no surprise that they can be written in a convenient four-vector notation. Indeed, it was in Einstein's original relativity paper [7] that the covariant nature of Maxwell theory was first explicitly shown, although not in four-vector notation. Interestingly, most introductory-level physics courses, indeed most undergraduate physics courses, do not study the historical approach taken to the subject. Typically, the polished theory is shown on the first day (e.g. Newtonian Mechanics, Shrödinger equation), and the rest of the course is a working out of the consequences of the theory. There is a good reason for this; especially in more modern theories of physics, there are usually many different issues leading to the development of some heuristic rules which are only later synthesized into a meaningful view of nature, and many times there are simply incorrect assumptions made in making these rules that are no longer useful to show to undergraduates, or the mathematics is unnecessarily clunky. For example, Maxwell's equations are almost always expressed in vector notation, whereas this notation is absent from Maxwell's original 1865 paper [8] (Gibbs, Heaviside, and others were yet to develop vector calculus). An introductory Quantum Mechanics course will discuss the historical origin of the theory, but certainly won't reproduce the Planck distribution, expound fully the Bohr-Sommerfeld quantization, or develop matrix mechanics, even though all of these things historically preceded the Shrödinger equation. Unfortunately, this approach typically does not extend to relativity, and an introductory modern physics course will typically present the clear derivation of the Lorentz transoformations from a "light-clock", essentially the way Einstein did it in his famous paper [7]. This is a physically transparent way of deriving them; however, with the introduction in 1907 of the so-called Minkowski spacetime [9], it has become easier to express the laws of mechanics and electrodynamics, indeed any theory, in a covariant form. We will briefly discuss special relativity in Section 2, then present the covariant Maxwell equations and some applications, and then discuss the generalization to curved spacetimes. Throughout, Latin indices will run over the spatial dimensions and Greek indices will run over the spacetime dimensions (i.e. $i, j, k... = 1, 2, 3$, $\mu, \nu, \delta... = 0, 1, 2, 3$) and the summation convention is in place.

II. SPECIAL RELATIVITY

Minkowski spacetime is a four dimensional inner product space with metric

$$\eta^{\mu\nu} = \eta_{\mu\nu} \equiv \mathrm{diag}[-1, 1, 1, 1]. \tag{1}$$

an element of this space is called a *four-displacement*,

$$x^\mu \equiv (ct, x^i) \tag{2}$$

where c is the speed of light, t is the "coordinate time" and x^i are the usual components of the spatial displacement. We can interpret

$$d\tau^2 = -\eta_{\mu\nu} dx^\mu dx^\nu \tag{3}$$

as the time elapsed on a clock after a small four-displacement (note that for a light ray, $|dx^i| = |dt| \implies d\tau = 0$, justifying the name given to lightlike curves in general relativity, "null curves"). Einstein realized this was the invariant entity; if one coordinate system x^μ and another coordinate system x'^μ both measure a proper time, they must agree, i.e.

$$\eta_{\mu\nu} dx^\mu dx^\nu = \eta_{\mu\nu} dx'^\mu dx'^\nu. \tag{4}$$

Any such coordinate transformation must be linear and therefore of the form $y = mx + b$,

$$x^\mu = \Lambda^\mu{}_\nu x'^\nu + a^\mu. \tag{5}$$

These are the Lorentz transformations. In two spacetime dimensions, neglecting the intercept, they take the form shown in a typical modern physics course,

$$t' = \gamma(t - |v^i|x) \tag{6}$$

$$x' = \gamma(x - |v^i|t) \tag{7}$$

where $\gamma \equiv \frac{dt}{d\tau} = (1 - v^2)^{-\frac{1}{2}}$ is the Lorentz factor and $v^i \equiv \frac{dx^i}{dt}$ is the three-velocity. Here the magnitude is defined as $|v^i| \equiv \sqrt{v^i \cdot v^i}$ with the dot product defined as the usual Euclidean inner product. Note that $|v^i| \leq 1$. These transformations form a group, the Poincaré or inhomogenous Lorentz group. Properties of this group and its representations play a fundamental role in the mathematical formulation of quantum field theory, but a lengthy discussion of this would take us too far afield here. It will suffice mention some of the physical ramifications of the Lorentz transformations for our discussion of mechanics and electrodynamics. First, note that the transformations mix time and space coordinates, leading to the well known time dilation and Fitzgerald-Lorentz length contraction. We define a four-velocity as

$$U^\mu \equiv \frac{dx^\mu}{d\tau} \tag{8}$$

and the "four-acceleration"

$$\alpha^\mu \equiv \frac{d^2x^\mu}{d\tau^2}. \tag{9}$$

Newton's second law may be taken as the definition of relativistic force

$$f^\mu \equiv m\alpha^\mu. \tag{10}$$

where m is the (rest) mass. This suggests defining p^μ as the four-momentum, with

$$p^i = \gamma m v^i \qquad p^0 = E = m\gamma = \sqrt{|p^i|^2 + m^2} \tag{11}$$

where E is the energy. Taylor expansion around $v << 1$ (the classical limit) yields

$$p^i \approx mv^i + O((v^i)^3) \qquad p^0 \approx m + \frac{m|v^i|^2}{2} + O(|v^i|^4) \tag{12}$$

yielding the expected results plus the most famous equation in physics. For a complete discussion of the conservation of these quantities, the reader is advised to see any book on special relativity, such as (Weinberg, MTW, Rindler, Sard).

Many physical systems are composed of systems without well defined particle four-momenta, such as a simple fluid or any field. For such systems, it is necessary to introduce an object that characterizes the density and the current of the four-momentum. By analogy to fluid dynamics, we assume that the matrix will be a generalization of the Cauchy stress tensor, although since Minkowski spacetime is isomorphic to $\mathbb{R}^4$ we expect it to be a symmetric 4×4 matrix. We define the *stress-energy-momentum tensor* as a symmetric rank 2 tensor

$$T^{\mu\nu}(x^\beta) \equiv \Sigma_n \int d\tau p^\mu \frac{dx^\nu}{d\tau} \delta^4(x^\beta - x_n^\beta). \tag{13}$$

From (11), we see that $p^i = Ev^i$, so we can write this tensor in a way that more clearly demonstrates that it is symmetric,

$$T^{\mu\nu}(x^\beta) \equiv \Sigma_n \int d\tau \frac{p^\mu p^\nu}{E} \delta^4(x^\beta - x_n^\beta) \tag{14}$$

If there are no forces acting on the system, then the conservation law takes the form

$$\frac{\partial}{\partial x^\mu} T^{\mu\nu}(x^\beta) = 0 \tag{15}$$

. (Note: This conservation law, generalized to arbitrary spacetimes, only applies to one tensor built out of the metric tensor and its first and second derivatives, the Einstein tensor, strongly suggesting Einstein's equation $G_{\mu\nu} = T_{\mu\nu}$.) Generally, the divergence of the stress energy tensor of a system subjected to an external force is

$$\frac{\partial}{\partial x^\mu} T^{\mu\nu} = G^\nu \tag{16}$$

where G^ν is the density of force, the force per unit volume.

III. ELECTROMAGNETISM

The canonical form of Maxwell's equations are (in Heaviside-Lorentz units, with c=1),

$$\oint\!\!\!\oint \vec{E} \cdot d\vec{A} = Q_{enc} \tag{17}$$

$$\oint\!\!\!\oint \vec{B} \cdot d\vec{A} = 0 \tag{18}$$

$$\oint \vec{E} \cdot d\vec{l} = -\frac{\partial}{\partial t} \oint\!\!\!\oint \vec{B} \cdot d\vec{A} \tag{19}$$

$$\oint \vec{B} \cdot d\vec{l} = \vec{I}_{enc} + \frac{\partial}{\partial t} \oiint \vec{E} \cdot d\vec{A}. \tag{20}$$

Two cases of the generalized Stokes' theorem, Gauss's divergence theorem as well as Stokes' theorem, can be used to convert these integral equations into differential equations (Divergence theorem on the first two, Stokes' theorem on the second two),

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \tag{21}$$

$$\vec{\nabla} \cdot \vec{B} = 0 \tag{22}$$

$$\vec{\nabla} \times \vec{E} = \frac{\partial \vec{B}}{\partial t} \tag{23}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \tag{24}$$

We see that in the static limit ($\frac{\partial}{\partial t}x = 0 \forall x$), we recover electrostatics through equation (16). As they stand, only to the trained eye are these equations Lorentz invariant. To demonstrate this more clearly, it is advisable to define the *field strength tensor* as

$$F^{i0} = -E^i \qquad F^{ij} = \epsilon^{ijk} B_k \qquad F^{\mu\nu} = -F^{\nu\mu} \tag{25}$$

The homogenous equations (Gauss's law for magnetism and Faraday's law) can be combined in the so-called "Gauss-Faraday law" [6]

$$\epsilon^{\alpha\beta\mu\nu} \frac{\partial}{\partial x^\beta} F^{\mu\nu} = 0. \tag{26}$$

This is merely a consequence of the field strength tensor satisfying a Jacobi identity

$$\frac{\partial}{\partial x^\alpha} F^{\mu\nu} + \frac{\partial}{\partial x^\mu} F^{\nu\alpha} + \frac{\partial}{\partial x^\nu} F^{\alpha\mu} = 0. \tag{27}$$

For the inhomogenous equations (Gauss's Law for electric fields and Ampere's law with Maxwell's correction), we first define the *current density four-vector*

$$J^\mu = (\rho, j^i) \tag{28}$$

where ρ is the charge density and $j^i = \rho v^i$ is the three-current density. We see that

$$\frac{\partial}{\partial x^\mu} F^{\mu\nu} = -J^\nu, \tag{29}$$

an expression known as the Gauss-Ampere law. The Lorentz force law

$$\frac{dp^i}{dt} = q[E^i + v^i \times B^i] \tag{30}$$

can be written as

$$\frac{dp^\mu}{d\tau} = qF^\mu{}_\nu U^\nu. \tag{31}$$

Before moving on to the stress-energy-momentum tensor for the electromagnetic field, we note that because $F^{\mu\nu}$ satisfies the Jacobi identity, we can write it as the curl of a four-vector

$$F^{\mu\nu} = \frac{\partial}{\partial x_\mu} A^\nu - \frac{\partial}{\partial x_\nu} A^\mu. \tag{32}$$

The four-potential A is called the gauge field of the theory, for reasons which will be clear later.

IV. ELECTROMAGNETIC STRESS ENERGY TENSOR AND LAGRANGIAN

When particles are subjected to forces that are not body forces, such as the Lorentz force, equation (16) reads

$$\frac{\partial}{\partial x^\mu} T^{\mu\nu} = \Sigma_n q_n F^\mu{}_\nu \frac{dx_n^\nu}{dt} \delta^3(\vec{x^i} - \vec{x_n^i}) = G_L^\mu \tag{33}$$

where G_L^μ is the Lorentz force per unit volume. Note that this is equivalent to

$$\frac{\partial}{\partial x^\mu} T^{\mu\nu} = F^\mu{}_\nu J^\nu \tag{34}$$

Thus, we can construct a conserved electromagnetic stress energy tensor by adding a term that satisfies

$$\frac{\partial}{\partial x^\mu} T_{EM}^{\mu\nu} = -F^\mu{}_\nu j^\nu \tag{35}$$

This tensor turns out to be the spacetime generalization of the *Maxwell Stress Tensor*. This tensor can be found in Griffith's *Electrodynamics*[10]. In our units,

$$\sigma_{ij} \equiv \frac{1}{4\pi}(E_i E_j + B_i B_j - \frac{1}{2}(E_i E + B_i B^i)\delta_{ij}) \tag{36}$$

It is obviously symmetric, as (14) tells us it must be if it will be the spacelike parts of the *Electromagnetic stress-energy tensor*, with timelike components the well known electromagnetic energy density and Poynting vector

$$T^{00} = \frac{1}{2}(E_i E^i + B_i B^i) \qquad T^{0i} = \epsilon^{ijk} E_j B_k. \tag{37}$$

More compactly [11],

$$T_{EM}^{\mu\nu} = (F^{\mu\gamma} F^\nu{}_\gamma - \frac{1}{4}\eta^{\mu\nu} F_{\gamma\delta} F^{\gamma\delta}) \tag{38}$$

Thus,

$$T^{\mu\nu} \equiv \Sigma_n q_n F^{\mu}{}_{\nu} \frac{dx_n^{\nu}}{dt} \delta^3(\vec{x^i} - \vec{x^i_n}) + (F^{\mu\gamma} F^{\nu}{}_{\gamma} - \frac{1}{4} \eta^{\mu\nu} F_{\gamma\delta} F^{\gamma\delta}) \tag{39}$$

is our new stress-energy tensor and is conserved, $\partial_\mu T^{\mu\nu} = 0$.

The last tool we will need to study the gauge invariance of the electromagnetic field is the Lagrangian. Maxwells equations should satisfy an *action principle*, the statement that the action functional is a stationary point[1].

$$\frac{\delta}{\delta \psi_i} S = \frac{\delta}{\delta \psi_i} \int \mathcal{L} \, d^4x \tag{40}$$

where $\mathcal{L}$ is the Lagrangian density. Integration by parts, as well as the requirement that the action be a stationary point at the boundary of integration, yields the requirement (see [1]),

$$\frac{\partial \mathcal{L}}{\partial \psi_i} + \partial_\mu (\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi_i)}) = 0 \tag{41}$$

where $\frac{\partial}{\partial x^\mu} \equiv \partial_\mu$ is the four-gradient. Thus, varying the action w.r.t. the potential field, (replacing ψ_i above with A_μ),

$$\mathcal{L} = \frac{-1}{4} F^{\mu\nu} F_{\mu\nu} + A^\mu J_\mu \tag{42}$$

is a suitable Lagrangian that reproduces the Gauss-Ampere law above.

V. GAUGE INVARIANCE

Note that transformations of the type

$$A'_\mu = A_\mu + \partial_\mu \lambda \tag{43}$$

leave the $F^{\mu\nu}$ invariant by virtue of equation (32). Such a transformation is called a *gauge transformation*, and this invariance under transformations which potentially depend on the position in space-time is a type of *gauge invariance*. Because Maxwell's theory is invariant under such a transformation it is known as a *gauge theory* or *Yang-Mills Theory*(Cite Yang Mills here). This allows us to use the *Lorenz Gauge Condition*(Griffith's particles),

$$\partial_\mu A^\mu = 0 \tag{44}$$

Maxwells equations then take the form

$$\Box^2 A^\mu = -J^\mu \tag{45}$$

where $\Box^2 \equiv \partial^\mu \partial_\mu$ is the relativistic Laplace operator, or the *d'Alembertian*. In vacuum, this becomes a wave equation,

$$\Box^2 A^\mu = 0 \tag{46}$$

with the speed of the wave equal to the speed of light (1 in our units). There are of course other choices we could make for our gauge, another popular one being the *Coulomb gauge*, "which spoils the manifest Lorentz covariance of the theory"(Griffith's particles)

$$A^0 = 0 \implies \mathrm{Div}(A^i) = 0. \tag{47}$$

There is much more to discuss regarding gauge invariance. Truly, it is a statement about the lagrangian of the theory being invariant under a transformation. The general Lagrangian of a spin-1 quantum field can be found in any textbook on Quantum field theory (QFTFTGA, Weinberg, Peskin and Shroeder). It is the *Proca Lagrangian*,

$$\mathcal{L} = \frac{-1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{1}{2} (\frac{mc}{\hbar})^2 A^\mu A_\mu \tag{48}$$

Note that the first term is invariant under the gauge transformation (43), but the second term certainly is not. This should come as no surprise; as Jackson points out in his introduction [3],

> "The distance dependence of the electrostatic law of force was shown quantitatively by Cavendish and Coulomb to be an inverse square law."

By definition, the gradient of the potential is proportional to the force, so we expect our potential to look like the Yukawa potential

$$\Phi(r) = \frac{e^{-\frac{m}{\hbar} r}}{r} \tag{49}$$

in the limit $m \to 0$. In this limit, the Proca equation becomes the free-field Maxwell Lagrangian.

VI. CURVED SPACETIMES

The spacetime of Minkowski possesses ten symmetries of the metric. Four of these can be seen to be results of displacement of a coordinate system, i.e. the four components of a^μ in equation (5). These are perhaps the four most familiar symmetries; their generators are the three components of the linear momentum and the Hamiltonian. However, Einstein completed in 1915 the first physical theory to take advantage of *non-euclidean geometry*. Euclid in his *Elements* implicitly assumed that spaces were symmetric, i.e. there exist killing vectors (see [2],[6],[11]). General Relativity instead has the metric as the key dynamical variable, representing the gravitational field (it is still symmetric, but the space it represents no longer necessarily is). Indeed, it is a subject of great debate which is the "real" entity: space-time or the gravitational field (see [2], [13])). The view held by Einstein and the author of this paper is that the "real" physical object is the gravitational field, and spacetime is merely another word for this field, which becomes a constant (the Minkowksi tensor) in the limit that the gravitational field is weak.

Surprisingly, formulating equations in general relativity does not require the use of the Einstein equation. Indeed, *general covariance*, which can be viewed as a generalization of the equivalence principle, states that two steps must be taken to generalize a flat space-time equation to general space-times. One takes the metric $\eta_{\mu\nu}$ and replaces it with a general metric $g_{\mu\nu}$,[2] and replaces all of the four-gradients with *covariant derivatives* ∇_μ, which can be determined from the connection $\Gamma^\mu{}_{\nu\lambda}$, which is related to the metric (for a general vector A_μ, ($\nabla_\nu A_\mu = \partial_\nu A_\mu + \Gamma^\lambda{}_{\nu\mu} A_\lambda$)). Indeed, this is the same process used to construct gauge-invariant lagrangians in the Standard Model. Here, we merely state generally covariant forms of the definition of the Faraday tensor,

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\nu, \tag{50}$$

and the covariant Lorentz force constructed therefrom,

$$f^\mu = qF^\mu{}_\nu U^\nu, \tag{51}$$

where $F^\mu{}_\nu = g^{\mu\lambda} F_{\lambda\nu}$ (see [2],[6],[11]).

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[1] This point is technically an element in the function space, and the set of of functions which satisfy this constraint are known as the dynamical shell. Hamilton's principle function is then the on-shell action functional, see [12].

[2] $g_{\mu\nu}$ must satisfy the obvious mathematical requirements for a distance function, as well as whatever constraints are put on it by the Einstein equation).